Rain- Based Traffic Alert System

Team No: 6

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MileStone 2: Dataset Generation and ML model Development

1. Introduction

The second milestone focuses on building the data-driven component of the Intelligent Rain-Based Traffic Alert System. This includes generating a dataset from simulated sensor values and applying machine learning techniques to classify road safety conditions. The goal is to improve prediction accuracy and enable intelligent decision-making.

2. Dataset Generation

The dataset is synthetically generated using simulated sensor values obtained from the IoT system. Since real-time large-scale data collection is not feasible at this stage, random data generation is used to create multiple scenarios.

Features Used:

- Rain (0-4095) - Rain intensity
- Traffic (0 or 1) - Vehicle presence
- Distance (cm) - Distance between vehicles
- Temperature (°C) - Environmental condition
- Humidity(%) - Moisture level in air
- Visibility (low/medium/high or numeric)- Road visibility
- Speed(km/h) - Vehicle speed
- Risk - Output label (SAFE, CAUTION, HIGH, CRITICAL)

2.1 Dataset Creation Method

- Random values are generated for each feature
- Different combinations simulate real-world conditions

- Each data point is labeled based on predefined rules
- Dataset is stored in CSV format for ML processing

2.2 Head of Dataset

```
Total rows: 1500

Sample data:
   Rain  Traffic  Distance  Temperature  Humidity  Visibility  Speed  Risk
0   964        0       195           21        52         100     24  Safe
1  2582        1        94           35        41          16     27  High
2  3663        1        56           40        40          95     75  High
3  3464        0        64           22        95          28    100  High
4  3298        1       109           40        92          70     48  High
```

3. Risk Classification Logic

1. CRITICAL : It combines High rain , traffic present and very low distance
2. HIGH : Medium rain or moderate distance
3. CAUTION : Low rain and safe distance
4. SAFE : No traffic and normal conditions

4. Machine Learning Model

Models Used:

1.Decision Tree

- Rule-based classification
- Easy to interpret

2.Random Forest

- Ensemble of multiple trees
- High accuracy and robustness

3.K-Nearest Neighbors (KNN)

- Distance-based classification
- Works well for pattern similarity
- Final prediction based on majority voting

4. Logistic Regression

- Statistical model for classification
- Fast and efficient
- Works well with structured data

5. Support Vector Classifier (SVC)

- Separates data using optimal boundaries
- Performs well for complex classification problems

6.Hybrid Model (Voting Classifier)

- Combines logistic regression, SVC and Decision Tree Classifier.

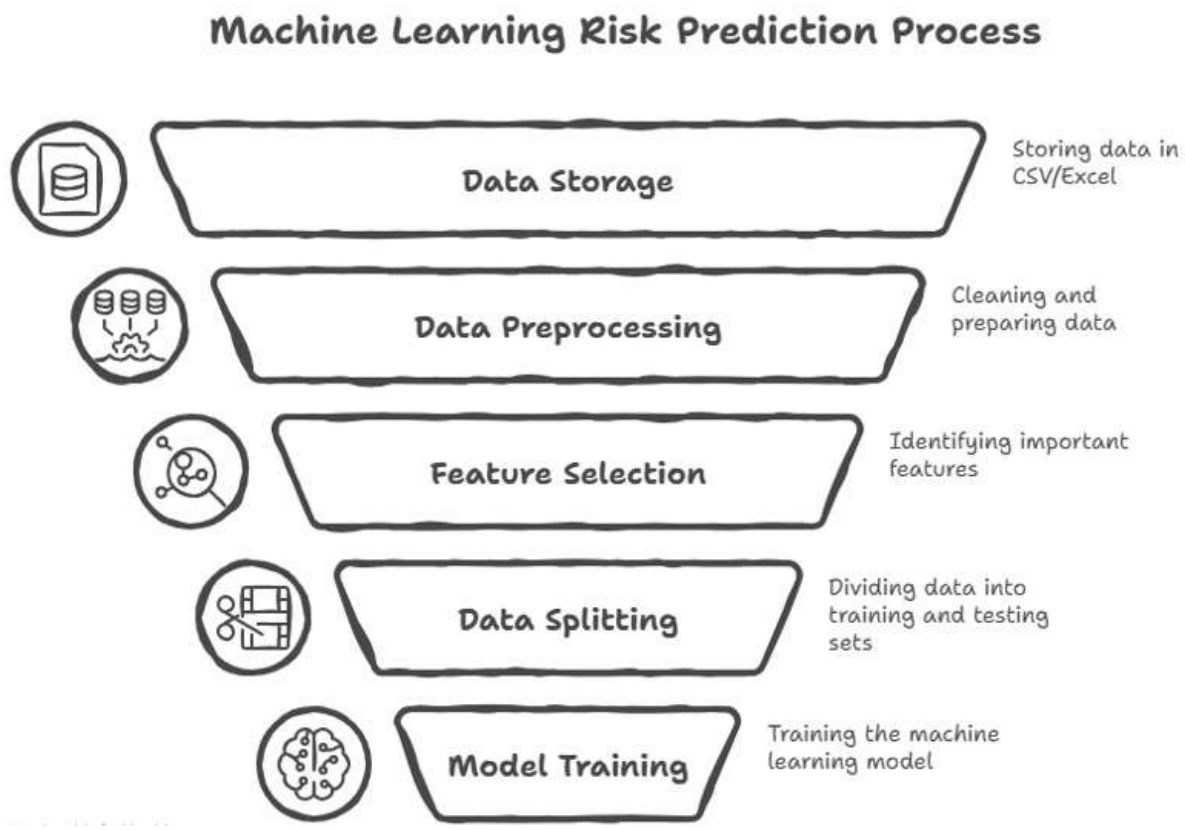
4.1 Model Workflow

1. Load dataset
2. Split into training and testing data
3. Train ML model
4. Predict risk levels
5. Evaluate accuracy

6. Data Storage in Excel

The generated dataset is stored in an Excel format to enable easy visualization, analysis, and sharing. Excel is used because it provides a simple tabular structure and supports integration with machine learning tools.

Workflow



7. Code

7.1 DataSet Creation and Storing in Excel Automatically

```
import pandas as pd
import random
num_rows = 1500
data = []
for i in range(num_rows):
    rain = random.randint(0, 4095)
    traffic = random.randint(0, 1)
    distance = random.randint(5, 200)
    temperature = random.randint(20, 40)
    humidity = random.randint(40, 100)
    visibility = random.randint(10, 100)
    speed = random.randint(20, 100)
```

```
    if rain > 3000 and distance < 20 and traffic == 1:
        risk = "Critical"
    elif rain > 2000 or distance < 40 or visibility < 30:
        risk = "High"
    elif rain > 1000 or speed > 80:
        risk = "Caution"
    else:
        risk = "Safe"
    data.append([
        rain, traffic, distance, temperature,
        humidity, visibility, speed, risk
    ])
df = pd.DataFrame(data, columns=[
    "Rain",
    "Traffic",
    "Distance",
    "Temperature",
    "Humidity",
    "Visibility",
    "Speed",
    "Risk"
])
df.to_excel("rain_traffic_dataset.xlsx", index=False)
df.to_csv("rain_traffic_dataset.csv", index=False)
print("Dataset successfully created!")
print("Total rows:", len(df))
print("\nSample data:")
print(df.head())
```

7.2 Random Forest Model Training

```
import pandas as pd
from sklearn.model_selection import train_test_split
from sklearn.ensemble import RandomForestClassifier
from sklearn.metrics import accuracy_score, classification_report, confusion_matrix
from sklearn.preprocessing import LabelEncoder
import joblib
df = pd.read_excel("rain_traffic_dataset.xlsx")
print("Dataset Loaded!")
print(df.head())
le = LabelEncoder()
df['Risk'] = le.fit_transform(df['Risk'])
print("\nRisk Classes:", list(le.classes_))
X = df[['Rain', 'Traffic', 'Distance', 'Temperature',
        'Humidity', 'Visibility', 'Speed']]
y = df['Risk']
X_train, X_test, y_train, y_test = train_test_split(
    X, y, test_size=0.2, random_state=42, stratify=y
)
model = RandomForestClassifier(
    n_estimators=300,
    max_depth=15,
    min_samples_split=2,
    min_samples_leaf=1,
    random_state=42
)
model.fit(X_train, y_train)
y_pred = model.predict(X_test)
accuracy = accuracy_score(y_test, y_pred)
```

```
print("\n MODEL TRAINED")
print("Accuracy: {:.2f}%".format(accuracy * 100))
print("\nClassification Report:")
print(classification_report(y_test, y_pred))
print("\nConfusion Matrix:")
print(confusion_matrix(y_test, y_pred))
joblib.dump(model, "high_accuracy_model.pkl")
print("\nModel saved as high_accuracy_model.pkl")
sample = [[3500, 1, 10, 30, 80, 20, 90]]
prediction = model.predict(sample)
print("\nSample Prediction (Encoded):", prediction)
print("Predicted Risk:", le.inverse_transform(prediction))
```

7.3 Other Models Training

```
from sklearn.svm import SVC
from sklearn.linear_model import LogisticRegression
from sklearn.neighbors import KNeighborsClassifier
from sklearn.tree import DecisionTreeClassifier
models = {
    "SVM": SVC(),
    "Logistic Regression": LogisticRegression(max_iter=1000),
    "KNN": KNeighborsClassifier(n_neighbors=5),
    "Decision Tree": DecisionTreeClassifier(max_depth=5)
}
for name, model in models.items():
    model.fit(X_train, y_train)
    y_pred = model.predict(X_test)
    acc = accuracy_score(y_test, y_pred)
```

```
print(f"\n{name} MODEL")
print("Accuracy:", acc * 100)
print(confusion_matrix(y_test, y_pred))
```

7.4 Results and Output

```
Dataset Loaded!
   Rain  Traffic  Distance  Temperature  Humidity  Visibility  Speed  Risk
0   964        0       195           21        52         100     24  Safe
1  2582        1        94           35        41          16     27  High
2  3663        1        56           40        40          95     75  High
3  3464        0        64           22        95          28    100  High
4  3298        1       109           40        92          70     48  High

Risk Classes: ['Caution', 'Critical', 'High', 'Safe']

 MODEL TRAINED
Accuracy: 99.33%

Classification Report:
              precision    recall  f1-score   support

           0       1.00      1.00      1.00        55
           1       1.00      0.33      0.50         3
           2       0.99      1.00      1.00       203
           3       1.00      1.00      1.00        39

    accuracy                           0.99       300
   macro avg       1.00      0.83      0.87       300
weighted avg       0.99      0.99      0.99       300
...
Model saved as high_accuracy_model.pkl

Sample Prediction (Encoded): [1]
Predicted Risk: ['Critical']
```

Fig7.3.1 Random Forest Accuracy

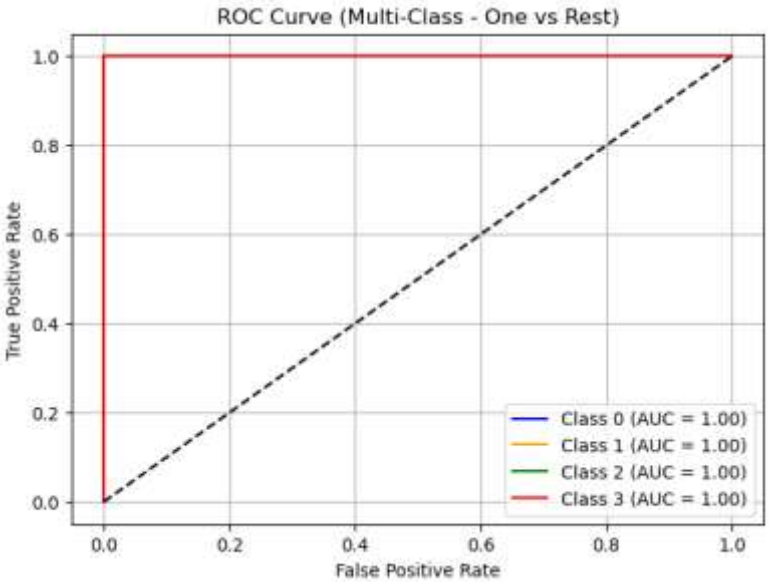


Fig7.3.2 Random Forest Curve

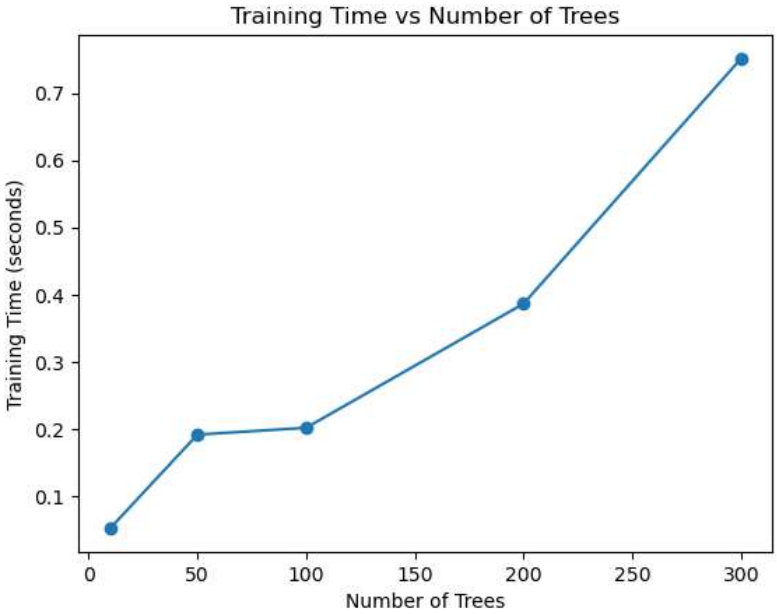


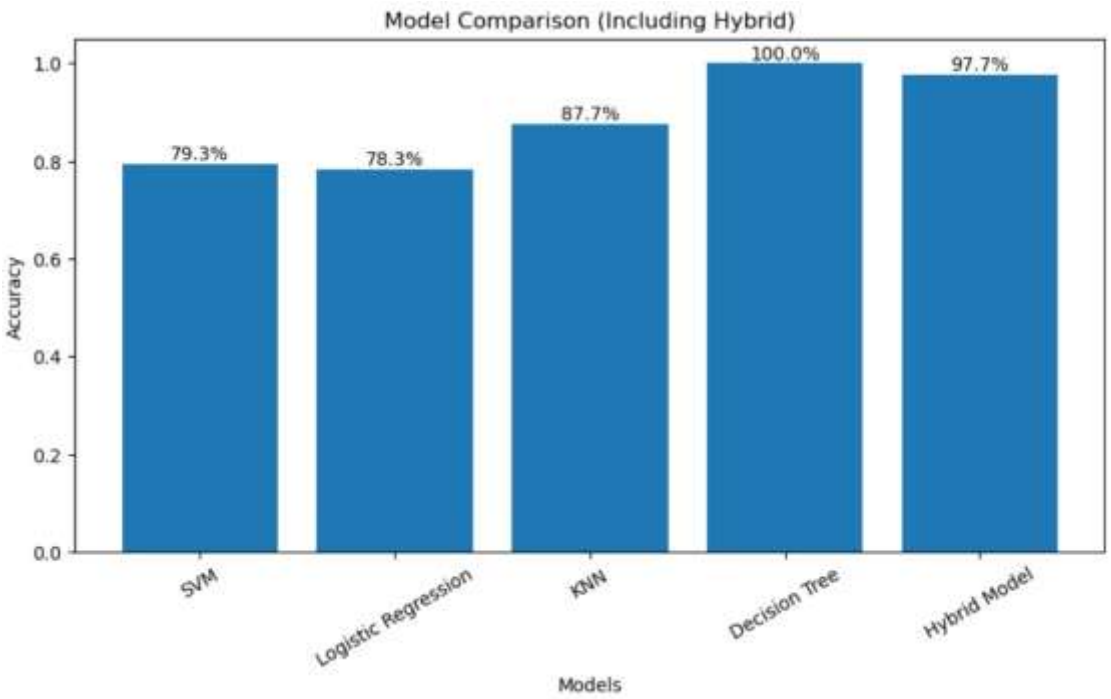
Fig7.3.3 Random Forest Training time

```
SVM MODEL
Accuracy: 79.33333333333333
[[ 42   0   8   5]
 [  0   0   3   0]
 [ 23   0 159  21]
 [  2   0   0  37]]

Logistic Regression MODEL
Accuracy: 78.33333333333333
[[ 25   0  27   3]
 [  0   1   2   0]
 [ 15   0 181   7]
 [  7   0   4  28]]

KNN MODEL
Accuracy: 87.66666666666667
[[ 50   0   1   4]
 [  0   0   3   0]
 [ 16   0 181   6]
 [  0   0   7  32]]

Decision Tree MODEL
Accuracy: 100.0
[[ 55   0   0   0]
 [  0   3   0   0]
 [  0   0 203   0]
 [  0   0   0  39]]
```

Fig7.3.4 Accuracy and Confusion matrix of different models

6. Comparative Study

The comparative study of different machine learning models highlights variations in their performance for traffic risk prediction. Support Vector Machine (SVM) and Logistic Regression achieved accuracies of 79.3% and 78.3% respectively, indicating moderate performance in handling the dataset. K-Nearest Neighbors (KNN) performed better with an accuracy of 87.7%, showing its effectiveness in identifying patterns based on similarity. The Decision Tree model achieved the highest accuracy of 100%, demonstrating its strong capability in rule-based classification. The Hybrid Model, which combines predictions from multiple models, achieved an accuracy of 97.7%, showing improved performance compared to most individual models.



6. Conclusion

The Intelligent Rain-Based Traffic Alert System demonstrates how IoT and machine learning can be combined to improve road safety during adverse weather conditions. The system uses simulated sensor data such as rain intensity, traffic presence, vehicle distance, temperature, humidity, visibility, and speed, which is stored in an Excel format and used to create a dataset for analysis and model training. Multiple machine learning models, including Support Vector Machine (79.3%), Logistic Regression (78.3%), K-Nearest Neighbors (87.7%), Decision Tree (100%), and a Hybrid Model (97.7%), were implemented and compared. The results show that while individual models perform well, the hybrid model provides more consistent and reliable predictions by combining the strengths of multiple algorithms. The system successfully classifies road conditions into SAFE, CAUTION, HIGH, and CRITICAL levels, helping in early risk detection.